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1. REVIEW: CYCLE NOTATION

Recall that S_n is the group of all bijections of $[n] = \{1, 2, \dots, n\}$, with composition as its group operation.

Definition 1.1 (Cycle). For distinct elements $i_1, \dots, i_k \in [n]$, the *cycle*

$$(i_1 \ i_2 \ \cdots \ i_k) \tag{1.1}$$

is the permutation sending i_1 to i_2 , i_2 to i_3 , and so on, with i_k sent back to i_1 . It fixes every element not listed in the cycle.

Example 1.1. The cycle $\sigma = (5 \ 3 \ 2) \in S_7$ is the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 5 & 2 & 4 & 3 & 6 & 7 \end{pmatrix}. \tag{1.2}$$

Thus $5 \mapsto 3 \mapsto 2 \mapsto 5$, while $1, 4, 6, 7$ are fixed.

If $\sigma, \tau \in S_n$, then $\sigma\tau = \sigma \circ \tau$ means that we apply τ first and then σ .

Remark. The inverse of a cycle is again a cycle: it traverses the entries in the opposite direction. Also, if $\sigma = (i_1 \ \cdots \ i_k)$ and $\tau \in S_n$, then

$$\tau^{-1}\sigma\tau = (\tau^{-1}(i_1) \ \tau^{-1}(i_2) \ \cdots \ \tau^{-1}(i_k)). \tag{1.3}$$

In particular, conjugating a cycle gives another cycle.

2. SUBGROUPS

Definition 2.1 (Subgroup). Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is a *subgroup* if it satisfies the following conditions:

- (1) $e \in H$.
- (2) H is closed under products: if $h_1, h_2 \in H$, then $h_1 h_2 \in H$.
- (3) H is closed under inverses: if $h \in H$, then $h^{-1} \in H$, where the inverse is taken in G .

A subgroup is itself a group under the same operation, with the same identity as G . Associativity is inherited from G .

Proposition 2.1 (Finite subgroup criterion). *If $H \subseteq G$ is finite, contains e , and is closed under products, then H is a subgroup. In this case, closure under inverses follows from the other two conditions.*

Proof. Let $a \in H$. Closure under products implies that

$$a, a^2, a^3, a^4, \dots \in H. \quad (2.1)$$

Since H is finite, there must be a repetition, say $a^r = a^s$ with $r > s > 0$. Multiplying by a^{-s} in G gives $a^{r-s} = e$. Consequently,

$$a a^{r-s-1} = a^{r-s-1} a = e. \quad (2.2)$$

The exponent $r - s - 1$ is nonnegative, so $a^{r-s-1} \in H$: if the exponent is zero, this uses $e \in H$. Thus $a^{-1} = a^{r-s-1} \in H$. $\square$

Remark. For an infinite subset, closure under inverses must still be checked. For example, $\{0, 1, 2, \dots\} \subseteq (\mathbb{Z}, +, 0)$ contains the identity and is closed under addition, but it is not closed under additive inverses.

3. GENERATED SUBGROUPS

Definition 3.1 (Cyclic subgroup). Let G be a group, with identity denoted by 1, and let $x \in G$. The *cyclic subgroup generated by x* is

$$\langle x \rangle = \{x^m : m \in \mathbb{Z}\} = \{1, x^{\pm 1}, x^{\pm 2}, x^{\pm 3}, \dots\}. \quad (3.1)$$

This is a subgroup: it contains $x^0 = 1$, products are given by $x^m x^n = x^{m+n}$, and $(x^m)^{-1} = x^{-m}$.

Definition 3.2 (Subgroup generated by a subset). For a subset $W \subseteq G$, let $\langle W \rangle$ be the set of all finite products

$$w_1 w_2 \cdots w_k, \quad k \geq 0, \quad (3.2)$$

where each w_i belongs to W or has its inverse in W . When $k = 0$, the product is the identity 1. This is called the *subgroup generated by W* .

Indeed, concatenating two such products gives another such product, and

$$(w_1 w_2 \cdots w_k)^{-1} = w_k^{-1} \cdots w_2^{-1} w_1^{-1} \quad (3.3)$$

is again of the required form. Hence $\langle W \rangle$ is a subgroup.

Example 3.1 (Subgroups of the integers). In the additive group $(\mathbb{Z}, +, 0)$, the even integers, $\{0\}$, and $\mathbb{Z}$ are all subgroups. More generally, for $a \in \mathbb{Z}$,

$$\mathbb{Z}a = \langle a \rangle = \{ma : m \in \mathbb{Z}\} = \{0, \pm a, \pm 2a, \pm 3a, \dots\} \quad (3.4)$$

is the subgroup consisting of all multiples of a . Here the operation is addition, so powers in multiplicative notation become integer multiples in additive notation.

Remark. For any group G , the set $\{e\}$ is a subgroup, called the *trivial subgroup*. The whole group G is also a subgroup of itself.

4. INTERSECTIONS OF SUBGROUPS

Proposition 4.1. *If H and K are subgroups of G , then $H \cap K$ is a subgroup of G .*

Proof. Since $e \in H$ and $e \in K$, we have $e \in H \cap K$. If $a, b \in H \cap K$, then $ab \in H$ and $ab \in K$, because both subgroups are closed under products. Thus $ab \in H \cap K$. Similarly, if $a \in H \cap K$, then $a^{-1} \in H$ and $a^{-1} \in K$, so $a^{-1} \in H \cap K$. $\square$

Remark. The union $H \cup K$ need not be a subgroup. For example, $2\mathbb{Z} \cup 3\mathbb{Z}$ contains 2 and 3, but does not contain their sum 5.

5. EVERY SUBGROUP OF THE INTEGERS IS CYCLIC

Lemma 5.1. *Every subgroup of $(\mathbb{Z}, +, 0)$ is either $\{0\}$ or $\mathbb{Z}d$ for some positive integer d .*

Proof. Let $H \subseteq \mathbb{Z}$ be a subgroup with $H \neq \{0\}$. Since H contains a nonzero integer and its negative, it contains a positive integer. By the well-ordering principle, H has a smallest positive element d . We claim that $H = \mathbb{Z}d$.

First, $\mathbb{Z}d \subseteq H$, since $d \in H$ and H is closed under addition and additive inverses.

For the reverse inclusion, let $m \in H$. By division with remainder, write

$$m = qd + r, \quad q \in \mathbb{Z}, \quad 0 \leq r < d. \quad (5.1)$$

Since $m, qd \in H$, we have $r = m - qd \in H$. If $r > 0$, this contradicts the choice of d as the smallest positive element of H . Thus $r = 0$, so $m = qd \in \mathbb{Z}d$. Hence $H \subseteq \mathbb{Z}d$, proving the claim. $\square$

6. GREATEST COMMON DIVISORS

Let $a, b \in \mathbb{Z}$. Consider the set of all integer linear combinations of a and b :

$$\mathbb{Z}a + \mathbb{Z}b = \{ma + nb : m, n \in \mathbb{Z}\}. \quad (6.1)$$

This is a subgroup of $(\mathbb{Z}, +, 0)$: it contains 0, and sums and negatives of integer linear combinations are again integer linear combinations. Therefore the preceding lemma gives

$$\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d \quad (6.2)$$

for some $d \geq 0$. If a and b are not both zero, then $d > 0$. We will show that this positive generator is $\gcd(a, b)$.

Lemma 6.1 (Divisibility and inclusion). *For integers r, s ,*

$$r \mid s \iff \mathbb{Z}s \subseteq \mathbb{Z}r. \quad (6.3)$$

Proof. If $s = kr$ for some $k \in \mathbb{Z}$, then every multiple of s is a multiple of r . Conversely, if $\mathbb{Z}s \subseteq \mathbb{Z}r$, then $s \in \mathbb{Z}r$, so $r \mid s$. $\square$

Proposition 6.1. *Let $a, b \in \mathbb{Z}$ be not both zero, and let $d > 0$ satisfy $\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d$. Then $d = \gcd(a, b)$. In particular,*

$$\mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}\gcd(a, b). \quad (6.4)$$

Proof. We check two properties:

- (1) d divides both a and b .
- (2) Every common divisor of a and b divides d .

For the first property, the inclusions

$$\mathbb{Z}a \subseteq \mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d, \quad \mathbb{Z}b \subseteq \mathbb{Z}a + \mathbb{Z}b = \mathbb{Z}d \quad (6.5)$$

imply $d \mid a$ and $d \mid b$ by Lemma 6.1.

For the second property, suppose $\ell \mid a$ and $\ell \mid b$. Then $\mathbb{Z}a \subseteq \mathbb{Z}\ell$ and $\mathbb{Z}b \subseteq \mathbb{Z}\ell$. Adding elements from these two subgroups gives

$$\mathbb{Z}d = \mathbb{Z}a + \mathbb{Z}b \subseteq \mathbb{Z}\ell. \quad (6.6)$$

By Lemma 6.1, $\ell \mid d$. Since $d > 0$, every positive common divisor is therefore at most d . Thus d is the greatest common divisor of a and b . $\square$